

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026 — Marking Report | Biology Class IX

Student: Seemab

Exam: exam-07-ch1-7 (Chapters 1 to 7 Comprehensive)

Attempt Date: 2026-06-23

Marked: 2026-06-23

57.5 / 6588.5%

Section A: 12/12 | Section B: 27.5/33 | Section C: 18/20
Biology subject avg: 91.5%

A

Correction issued 2026-06-23: this attempt was re-verified against the answer key. Q.3(ii) corrected from 3.5/5 to 3/5 because Chloroplast (worth 2 marks) was entirely omitted; exam total corrected from 58/65 to 57.5/65 (88.5%). The cumulative overall average is also corrected to 93.3%; earlier progress overviews carried an understated running total. Progress overview v17 is retained unchanged per the additive-only rule.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks) Score: 12/12

#	Question (shortened)	Seemab	Key	Result
1	Father of Taxonomy	C	C — Carolus Linnaeus	✓
2	Three-domain system proposed by	B	B — Carl Woese	✓
3	Powerhouse of the cell	D	D — Mitochondria	✓
4	Longest phase of mitosis	C	C — Prophase	✓
5	Levels smallest to largest	C	C — Cell, tissue, organ, organ system, organism	✓
6	Stable constant internal environment	C	C — Homeostasis	✓
7	Major bioelement constituting 99% protoplasm	B	B — Phosphorus	✓
8	Peptide bonds formed between	B	B — Amino acids	✓
9	Enzymes are globular proteins that	B	B — Tremendously speed up biochemical reactions	✓
10	Competitive inhibitor occupies	B	B — Active site	✓
11	ATP made up of Adenine, Ribose and	C	C — Three phosphate groups	✓
12	Vector for Malaria	C	C — Female Anopheles mosquito	✓

Perfect score — 12/12. Seventh consecutive perfect MCQ score (exams 14, 15, 16, 17 all 12/12 following end of the 11-exam streak).

SECTION B — Short Questions (11 × 3 = 33 Marks) Score: 27.5/33

Q.2(i) — Define Biology; state THREE major fields with one example organism each

2 / 3

Chose: Main question

Three major fields correctly named: Botany (plants, photosynthesis example), Zoology (animals, reproduction example), Microbiology (microorganisms, amoeba example). Fields component fully correct.
Deduction: No formal definition of Biology ("branch of science dealing with study of life and living organisms") was written. The definition carries 1 mark. Writing only the fields list without defining the subject first cost 1 mark.

Q.2(ii) OR — Role of Ronald Ross in proving Malaria is transmitted by Anopheles mosquito

3 / 3

Chose: OR variant

Excellent response. Correctly stated 1897, Plasmodium as causative agent, female Anopheles mosquito as vector, demonstrated parasite life cycle in mosquito gut, concept of vector defined, Nobel Prize in Medicine 1902 cited. All three marking points covered precisely.

Q.2(iii) OR — Three domains of Carl Woese; naming the basis of his classification

3 / 3

Chose: OR variant

Outstanding depth. Named all three domains (Archaea, Bacteria, Eukarya). Stated basis correctly: rRNA nucleotide sequence differences, plus cell membrane lipid structure and antibiotic sensitivity. Listed characteristics of each domain clearly. Carl Woese 1990 date correct. The answer goes beyond the minimum and demonstrates confident understanding of the topic.

Q.2(iv) OR — Compare SER and RER in structure and function

3 / 3

Chose: OR variant

Clean two-column table with three comparison rows. Ribosomes: present (rough appearance) vs absent. Function: protein synthesis and transport vs lipid synthesis and detoxification. Location: near nucleus vs throughout cytoplasm. All three marks secured.

Q.2(v) — Significance of mitosis under three headings: genetic stability, growth, wound healing

2 / 3

Chose: Main question

Genetic stability: well explained — daughter cells carry identical chromosomes, DNA replicated into exact copies beforehand, same genetic information maintained. Correct and detailed (1 mark). Development and growth: correctly noted that multicellular organisms start from a single zygote, grow by increasing cell number through mitosis (1 mark).
Deduction: Wound healing / cell replacement was not addressed at all. The question explicitly lists all three headings. Missing the third heading cost 1 mark. Seemab stopped after two sub-points. Drill fix: after completing a multi-heading answer, physically re-read the question and tick off every required heading before moving on.

Q.2(vi) OR — Distinguish between benign and malignant tumours; define metastasis

3 / 3

Chose: OR variant

Tumour correctly defined as abnormal mass from uncontrolled cell division. Benign: remains at original site. Malignant: cancerous, spreads to neighbouring and distant tissues. Metastasis defined precisely as "spread of cancer cells beyond their original site." All three marks awarded.

Q.2(vii) — How different tissues form the stomach; name four tissue types and roles

3 / 3

Chose: Main question

All four animal tissue types listed with accurate roles: Epithelial (inner lining, secretes mucus, HCl, pepsin enzyme), Connective (supports and connects mucosa to muscular layer), Muscle (contracts to move food through gut, peristalsis), Nervous (controls smooth muscle contraction). Full marks.

Q.2(viii) — Structure and function of three types of plant tissue

2.5 / 3

Chose: Main question

Dermal tissue: covers and protects the plant (1/1 mark). Vascular tissue: transports water, minerals, and sugars to different parts of plants. Sufficient for full credit (1/1 mark).

Deduction: Ground tissue described only as "serves as site for photosynthesis." The key additionally requires providing support and storing water and sugars. Half a mark deducted for incomplete function (0.5/1).

Q.2(ix) — Energy yield per gram of lipids, carbohydrates, proteins; THREE functions of lipids

0 / 3

Not attempted

This question was skipped entirely. The paper sequence goes from Q.2(viii) directly to Q.2(x) with no answer written for Q.2(ix). This cost 3 marks — the single largest deduction on the paper. The content (Lipids = 9.1 kcal/g, Carbohydrates = 4 kcal/g, Proteins = 4 kcal/g; functions: cell membrane component, energy storage, steroid hormones) is covered in Chapter 6 Molecular Biology. Q.2(ix) was confirmed not a missing page — a genuine skip. A time-management or oversight issue caused this omission.

Q.2(x) — Name three types of RNA; state specific role of each in protein synthesis

3 / 3

Chose: Main question

mRNA: carries DNA message from nucleus to cytoplasm for protein synthesis. tRNA: transfers amino acids to ribosomes during protein synthesis, anticodon matches codon. rRNA: component of ribosomes, combines with ribosomal proteins to form ribosome units. All three types with distinct roles fully stated.

Q.2(xi) OR — Name FOUR factors affecting enzyme activity; explain how temperature affects enzyme functioning

3 / 3

Chose: OR variant

Four factors correctly listed: Temperature, pH, Substrate concentration, Inhibitors/Activators. Temperature explanation: rate increases with temperature up to a specific limit, each enzyme has an optimum temperature, human enzyme optimum is 37°C. At very high temperatures atoms of enzyme molecules vibrate vigorously, globular structure and active site are damaged — enzyme is denatured. Precise, textbook-accurate explanation. Full marks.

SECTION C — Long Questions (4 × 5 = 20 Marks) Score: 18/20

Q.3(i) OR — Four advantages of classification; complete taxonomic hierarchy (domain to species) using human as example

5 / 5

Chose: OR variant

Four advantages stated clearly: makes study of organisms easy and systematic; helps understand evolutionary relationships; provides a universal language for scientists; facilitates identification of new species and organisms. All four correct (1 mark). Complete taxonomic hierarchy for human: Domain Eukarya, Kingdom Animalia, Phylum Chordata, Class Mammalia, Order Primates, Family Hominidae, Genus Homo, Species sapiens sapiens. All nine classification levels present and accurate (4 marks). Excellent recall under exam conditions.

Q.3(ii) — Structure and functions of: (a) Mitochondria, (b) Chloroplast, (c) Golgi apparatus

3 / 5

Chose: Main question

Mitochondria: structure described correctly (double membrane, outer smooth, inner folded into cristae, matrix, own DNA and ribosomes). Function: site of cellular respiration, produces ATP, 36 ATP per glucose in aerobic respiration. Excellent — 2 marks. Golgi apparatus: stack of flattened cisternae, discovered by Camillo Golgi. Receives proteins from RER, processes and packages into vesicles for secretion or delivery to lysosomes. 1 mark.

Deduction: Chloroplast was not addressed at all. The question explicitly lists (a), (b), and (c). Chloroplast carries 2 marks (structure: double membrane, thylakoids with grana, stroma; function: site of photosynthesis, converts light energy to chemical energy, contains chlorophyll). All 2 marks for Chloroplast are lost. This is the stopping-early pattern recurring in Section C. Drill fix: before starting Q.3, write out the list of all sub-parts from the question and tick each as you complete it.

Q.3(iii) OR — Explain the central dogma; describe transcription and translation in detail, naming all molecules involved

5 / 5

Chose: OR variant

Central dogma stated correctly: flow of genetic information from DNA to RNA to Proteins. Transcription: first step of gene expression, occurs in nucleus, both strands unwind, template strand is copied, mRNA formed, mRNA moves from nucleus to cytoplasm. Translation: occurs at ribosomes in cytoplasm, mRNA codons read by ribosomes, each codon specifies one amino acid, tRNA molecules carry specific amino acids with anticodon matching codon. All required molecules named: DNA, mRNA, tRNA, ribosomes, amino acids. One of the most complete Section C answers of all 17 attempts. Full marks.

Q.3(iv) OR — Define ATP and role as energy currency; ATP-ADP cycle; three components; high-energy bonds

5 / 5

Chose: OR variant

ATP as energy currency: all life processes depend on energy; living organisms store and use energy at cellular level as Adenosine Triphosphate. Three components: Adenine (double-ringed nitrogen base), Ribose (five-carbon sugar), Three phosphate groups (PO4) linked in triphosphate group. ATP-ADP cycle: Adenosine + 1P = AMP, Adenosine + 2P = ADP, Adenosine + 3P = ATP — correctly shown. High-energy bonds: covalent bonds between phosphate groups called high-energy bonds, represented by (~), store large amounts of energy. All four required elements addressed fully. Full marks.

Performance Summary

Strengths on this paper

Areas needing attention

• Perfect MCQs (12/12) covering all seven chapters — no conceptual gaps at the recall level. • Q.2(ix) skipped entirely (3 marks lost): No attempt made for energy yield per gram of

- Ronald Ross / Malaria answer (2(ii) OR): date, pathogen, vector, life cycle, Nobel Prize all cited from memory with marks-scheme precision.
- Three domains of Carl Woese (2(iii) OR): went well beyond the minimum, including distinguishing characteristics of each domain. The clearest Chapter 2 answer across all Biology exams so far.
- Central dogma (3(iii) OR): transcription and translation described with all required molecules, exactly matching the mark scheme. One of the top Section C performances across all 17 attempts.
- ATP structure and cycle (3(iv) OR): all four elements (energy currency role, three components, AMP/ADP/ATP progression, high-energy bonds) fully addressed.
- Taxonomic hierarchy (3(i) OR): complete and correct from Domain to Species with Homo sapiens sapiens.
- Mitochondria description: cristae, matrix, own DNA and ribosomes, 36 ATP in aerobic respiration — textbook accurate.
- lipids/carbs/proteins or functions of lipids. Before starting Section B, count how many questions are needed (11) and number each question slot to ensure none are accidentally skipped.
- **Q.3(ii): Chloroplast omitted (2 marks lost):** Three organelles were listed in the question — only two were answered. This is the third exam where a multi-part Section C sub-part is left blank. Drill fix: write out all sub-parts (a), (b), (c) before starting and tick each on completion.
- **Q.2(i): Missing Biology definition (1 mark lost):** The three fields were named correctly but the formal definition of Biology was not written. Multi-part questions must always address every stated requirement before listing sub-points.
- **Q.2(v): Wound healing omitted (1 mark lost):** Two of three required mitosis significance headings covered; wound healing/cell replacement not addressed. Recurring stopping-early pattern.
- **Q.2(viii): Ground tissue function incomplete (0.5 marks lost):** "Site for photosynthesis" correct but insufficient — support and storage of water/sugars also required by the mark scheme.

Biology Exam Comparison						
Exam	Date	MCQ	Section B	Section C	Total	%
exam-01-ch1-6	3 Apr 2026	11/12	22/33	19/20	52/65	80.0%
exam-02-ch7 (Apr)	30 Apr 2026	12/12	22.5/33	26/30*	60.5/65	93.1%
exam-02-ch7 (re-sit)	2 May 2026	12/12	33/33	20/20	65/65	100.0%
exam-03-ch4	8 Jun 2026	12/12	31/33	19.5/20	62.5/65	96.2%
exam-07-ch1-7	23 Jun 2026	12/12	27.5/33	18/20	57.5/65	88.5%

* exam-02-ch7 Apr had a different Section C structure; figures are approximate for comparison.